

COLES COUNTY AP, IL, USA

WMO: 725317

Lat: 39.478N

Lon: 88.280W

Elev: 220

StdP: 98.71

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.7	-20.7	0.6	-15.5	-18.0	0.8	-13.1	13.0	-4.5	11.9	-4.2	4.1	290	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	33.0	24.0	32.0	24.0	30.7	23.4	26.1	30.9	25.3	29.8	24.5	28.8	3.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.2	29.1	23.9	19.2	28.2	23.0	18.2	27.1	82.1	31.0	78.4	29.9	75.1	28.7	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.3	8.3	-20.3	35.3	3.4	2.6	-22.7	37.2	-24.7	38.8	-26.6	40.3	-29.0	42.2		
			-20.6	27.5	3.2	0.7	-23.0	28.0	-24.8	28.4	-26.7	28.8	-29.0	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-2.2	-0.1	5.9	12.5	18.1	22.7	24.2	23.4	20.0	13.3	6.6	0.5
	DBStd	10.62	6.66	6.46	6.23	5.21	4.63	3.27	3.14	2.84	4.06	5.17	5.59	6.12
	HDD10.0	1317	381	286	159	33	2	0	0	0	0	24	132	299
	HDD18.3	2901	637	515	387	184	64	6	1	2	28	171	354	552
	CDD10.0	2096	2	4	33	107	252	382	440	416	300	127	28	5
	CDD18.3	638	0	0	2	9	56	137	182	158	78	16	1	0
	CDH23.3	5599	0	0	7	64	422	1173	1743	1374	694	119	2	0
	CDH26.7	1887	0	0	0	5	94	392	660	475	234	26	0	0
Wind	WSAvg	3.9	4.7	4.8	4.7	4.7	3.9	3.2	2.8	2.7	3.0	3.8	4.3	4.5
Precipitation	PrecAvg	1058	61	55	86	101	114	100	115	84	85	83	96	79
	PrecMax	1438	196	143	196	219	294	265	264	272	225	237	303	224
	PrecMin	715	0	4	25	6	21	18	32	14	3	8	16	10
	PrecStd	171	41	34	46	54	60	57	65	53	60	52	57	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	18.5	24.4	27.4	31.0	33.7	36.7	34.3	33.4	29.4	22.8	17.7
		MCWB	14.1	14.9	17.2	18.7	22.3	23.1	24.8	24.7	22.0	21.4	15.8	15.2
	2%	DB	12.7	15.2	21.2	25.2	29.0	32.0	33.4	32.4	31.4	26.8	19.7	14.7
		MCWB	10.9	12.5	15.6	17.5	21.1	23.5	25.1	24.3	22.3	19.2	15.2	12.8
	5%	DB	10.1	12.4	18.4	23.0	27.5	30.7	32.0	31.0	29.5	24.0	17.4	11.9
		MCWB	8.2	9.6	13.7	16.6	20.1	23.1	25.0	23.6	21.6	17.6	13.6	10.2
	10%	DB	7.1	9.2	16.1	21.3	26.0	29.0	30.3	29.2	27.4	22.0	15.1	8.9
		MCWB	5.2	6.7	12.2	15.6	19.4	22.1	24.2	22.7	20.1	17.0	11.7	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.5	18.0	20.6	23.7	26.1	27.4	26.9	24.7	22.4	17.9	15.9
		MCDB	15.9	18.2	23.2	25.3	28.8	30.9	32.4	31.6	30.0	27.8	20.7	17.7
	2%	WB	11.6	12.9	16.4	19.0	22.5	25.0	26.5	25.7	23.6	21.0	15.9	13.0
		MCDB	12.6	14.6	20.3	23.3	27.3	29.6	31.5	30.4	28.7	24.5	18.3	14.4
	5%	WB	8.4	10.0	14.7	17.7	21.5	24.1	25.6	24.7	22.7	19.5	14.4	10.2
		MCDB	9.9	12.3	17.9	21.5	25.9	28.7	30.3	29.1	27.1	22.6	17.2	11.5
	10%	WB	5.1	6.9	12.5	16.4	20.4	23.3	24.9	23.9	21.7	17.5	12.0	7.3
		MCDB	7.1	9.1	15.6	20.3	24.4	27.6	29.2	27.9	25.6	20.7	14.8	8.6
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.6	9.9	10.9	10.3	10.1	10.1	10.6	11.9	10.7	9.1	7.4
		MCDBR	11.2	11.8	12.1	12.4	11.4	11.4	11.0	11.7	13.0	12.8	11.1	9.4
		MCWBR	9.8	9.6	7.9	7.2	5.7	5.1	4.7	5.2	5.3	6.7	7.8	8.1
	5% WB	MCDBR	10.9	11.3	10.9	11.0	10.2	9.7	9.7	10.0	10.5	10.5	10.3	8.9
		MCWBR	10.1	10.0	7.6	7.4	5.6	5.2	5.0	5.3	4.9	6.3	8.3	8.3
Clear-Sky Solar Irradiance	taub	0.303	0.320	0.353	0.398	0.435	0.436	0.460	0.460	0.416	0.361	0.315	0.301	
	taud	2.449	2.388	2.376	2.249	2.196	2.231	2.154	2.140	2.251	2.404	2.490	2.474	
	Ebn at Noon	869	900	902	878	848	845	820	809	824	840	849	844	
	Edh at Noon	80	98	110	134	144	139	149	147	124	96	77	73	
All-Sky Solar Radiation	RadAvg	1.80	2.65	3.66	4.79	5.45	6.29	6.13	5.52	4.64	3.26	2.18	1.53	
	RadStd	0.21	0.28	0.31	0.30	0.48	0.40	0.38	0.36	0.33	0.34	0.23	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+99	+70

Nomenclature: See separate page